F70LB Life Insurance Mathematics B

Heriot-Watt University

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1 Revision: Fundamentals of life insurance mathematics

We start by reviewing some of the most fundamental ideas from F70LA Life Insurance Mathematics A and other courses. This section recaps:

- The future lifetime random variable T_x, force of mortality μ_{x+t} and the probabilities ${}_tq_x = \mathrm{F}_x(t)$ and ${}_tp_x = \mathrm{S}_x(t)$
- The curtate lifetime random variable K_x
- The principle of equivalence and the expected present value of discrete or continuous payments that depend on death or survival

1.1 The future lifetime random variable

When a life takes out a policy, e.g. a life insurance policy which pays out a lump sum on death, the actual future lifetime is unknown.

We represent this future lifetime of a life aged x by a random variable T_x. Here are some of the important characteristics of T_x:

1. It is assumed to have a continuous distribution with cumulative distribution function (c.d.f.):

$$\mathrm{P}\{T_x \leq t\} = \mathrm{F}_x(t).$$

Associated with this is the survival function:

$$\mathrm{S}_x(t) = 1 - \mathrm{F}_x(t).$$

In actuarial notation:

$${}_tq_x = \mathrm{F}_x(t) \qquad {}_tp_x = \mathrm{S}_x(t)$$
$$\text{and} \quad {}_tq_x + {}_tp_x = 1.$$

2. T_x has probability density function (p.d.f.):

$$\begin{aligned} f_x(t) &= \frac{d}{dt}\mathrm{F}_x(t) = \frac{d}{dt}(1 - \mathrm{S}_x(t)) \\ &= -\frac{d}{dt}{}_tp_x. \end{aligned}$$

3. The force of mortality denoted μ_{x+t} is:

$$\begin{aligned}
\mu_{x+t} &= \lim_{h\to 0} \frac{\mathrm{P}\{T_x \le t+h \mid T_x > t\}}{h} \\
&= \lim_{h\to 0} \frac{\mathrm{F}_x(t+h) - \mathrm{F}_x(t)}{h\,(1-\mathrm{F}_x(t))} \\
&= \frac{1}{{}_tp_x} \lim_{h\to 0} \frac{\mathrm{F}_x(t+h) - \mathrm{F}_x(t)}{h} \\
&= \frac{1}{{}_tp_x} \frac{d}{dt}\mathrm{F}_x(t) \\
&= \frac{-\frac{d}{dt}{}_tp_x}{{}_tp_x}.
\end{aligned}$$

where, in the second line, we used the definition of a conditional probability to substitute:

$$\mathrm{P}\{T_x \le t+h \mid T_x > t\} = \frac{\mathrm{P}\{T_x \le t+h \text{ and } T_x > t\}}{\mathrm{P}\{T_x > t\}}$$

From $\mu_{x+t} = \frac{-\frac{d}{dt}{}_tp_x}{{}_tp_x}$, we have the Kolmogorov differential equation:

$$-f_x(t) = \frac{d}{dt}{}_tp_x = -{}_tp_x\mu_{x+t}$$

and since:

$$\frac{d}{dt}\log\left({}_tp_x\right) = \frac{\frac{d}{dt}{}_tp_x}{{}_tp_x}$$

we get:

$${}_tp_x = \exp\left(-\int_0^t \mu_{x+r}\; dr\right).$$

Note: You might or might not have seen the Kolmogorov differential equation above called by that name last semester. We call it that because it is a special case of the Chapman-Kolmogorov equations you learned about in the course Stochastic Processes. There is no need to worry about the connection for now. We will come back to that later this semester.

4. If we know the p.d.f. of T_x we can calculate moments of T_x:

$$\begin{aligned}
\mathrm{E}\left[T_x\right] &= \mathring{e}_x = \int_{t=0}^{\infty} {}_tp_x\; dt \\
\mathrm{var}\left[T_x\right] &= \mathrm{E}\left[T_x^2\right] - \left(\mathrm{E}\left[T_x\right]\right)^2.
\end{aligned}$$

The first of these, $\mathring{e}_x$, is the complete expection of life for a life currently aged x.

5. The probabilities ${}_tp_x$ are usually computed with the aid of a life table. A lifetable shows the notional number of people alive at each age if we started from an age α and a radix l_α. The radix represents the number of people alive at age α.

   We define the life table function l_x at all ages $x \ge \alpha$ by:

$$l_x = {}_{x-\alpha}p_\alpha\, l_\alpha.$$

   which represents the expected number of these same people who are still alive at age x.

   Therefore:

$${}_tp_x = \frac{l_{x+t}}{l_x}.$$

1.2 The curtate lifetime random variable

The random variable $K_x = \text{int}[T_x]$, the integer part of T_x, is the start of the year of death (measured from age x). The random variable K_x+1 is the end of the year of death. The same life table probabilities ${}_tp_x$, with integer values of t, serve to define the (discrete-valued) distribution of K_x and K_x+1. In fact, the term 'life table' usually refers to the tabulation of l_x at integer ages x.

1.3 EPVs of payments which depend on death or survival

Payments made on death, or as long as someone still lives, are made at random times, because T_x is a random variable. Given a force of interest δ, the present values of such payments are therefore also random variables. For example the present value of £1 payable immediately on death is the following (function of a) random variable:

$$e^{-\delta T_x} = v^{T_x}$$

Similarly, the present value of £1 payable at the end of the year of death is:

$$e^{-\delta(K_x+1)}$$

The expectations of such present values (EPVs) are important in pricing life insurance contracts. Many are given special symbols in the international actuarial notation. For example:

$$\begin{aligned} \bar{A}_x &= \mathrm{E}\left[e^{-\delta T_x}\right] \\ A_x &= \mathrm{E}\left[e^{-\delta(K_x+1))}\right] \\ \ddot{a}_x &= \mathrm{E}\left[\ddot{a}_{\overline{K_x+1}|}\right] \end{aligned}$$

Similar EPVs ($\bar{a}_x$, a_x, $\ddot{a}_x$, etc., may be defined in respect of level annuity payments. There are many variants of such symbols to deal with imited terms, deferred payment, frequency of payment and other simple variants.

The Principle of Equivalence is often used to calculate premiums. It states that:

$$\text{EPV of income} = \text{EPV of outgo}$$

so that, for example, the equation:

$$A_x = P_x \ddot{a}_x$$

is solved to find P_x, the premium for a whole-life assurance of £1, payable at the end of the year of death to a life age x, with premiums payable annually in advance for life.

(Note that the Principle of Equivalence is not true by definition. It just one way that an insurer could choose to set a premium. We will see others in this course.)

Standard EPVs can be calculated **exactly** for benefits payable at the end of the year of life, or annuities payable annually, using a standard life table, for example:

$$\begin{aligned} A_x &= \sum_{t=0}^{t=\infty} e^{-\delta(t+1)}\, {}_tp_x\, q_{x+t} \\ \ddot{a}_x &= \sum_{t=0}^{t=\infty} e^{-\delta t}\, {}_tp_x. \end{aligned}$$

When benefits are payable at the moment of death, or annuities are payable continuously, EPVs can be expressed **exactly** as integrals, for example:

$$\begin{aligned} \bar{A}_x &= \int_{t=0}^{t=\infty} e^{-\delta t}\, {}_tp_x\, \mu_{x+t}\, dt \\ \bar{a}_x &= \int_{t=0}^{t=\infty} e^{-\delta t}\, {}_tp_x\, dt \end{aligned}$$

but these integrals may have to be evaluated approximately using numerical methods. Such numerical methods (generally needing a computer) were not covered last semester but will be important this semester.